

Solve the Following System:

$$\begin{cases} \ln x + \ln y = 2\ln 3 + \ln 2 \\ e^{3x+1} \cdot e^{2-2y} = 1 \end{cases}$$

Solution:

$$\begin{aligned} \ln x + \ln y &= 2\ln 3 + \ln 2 \\ \Rightarrow \ln(xy) &= \ln(18) \\ e^{\ln(xy)} &= e^{\ln(18)} \\ \Rightarrow \boxed{xy = 18} \end{aligned}$$

$$\begin{cases} e^{3x+1} \cdot e^{2-2y} = 1 \\ e^{3x+1+2-2y} = 1 \\ e^0 = 1 \\ \text{So, } 3x+1+2-2y = 0 \\ \boxed{3x-2y+3=0} \end{cases}$$

$$\begin{cases} xy = 18 \\ 3x-2y+3=0 \end{cases}$$

$$\begin{aligned} \Rightarrow 3x &= -2y-3 \\ \boxed{x} &= \frac{-2y-3}{3} \end{aligned}$$

$$\Rightarrow \frac{(2y-3)}{3}(y) = 18$$

$$(2y-3)(y) = 54$$

$$2y^2 - 3y - 54 = 0$$

$$\Rightarrow y_1 = 6 \text{ and } y_2 = -4.5$$

$$\text{D.F. : } y > 0. \text{ so: } y = 6$$

$$\Rightarrow x = \frac{2(6) - 3}{3} = \frac{12 - 3}{3} = \frac{9}{3} = 3$$

$$\boxed{x = 3 \text{ and } y = 6}$$

$$S : (\{3, 6\})$$

